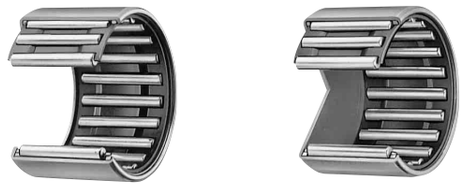
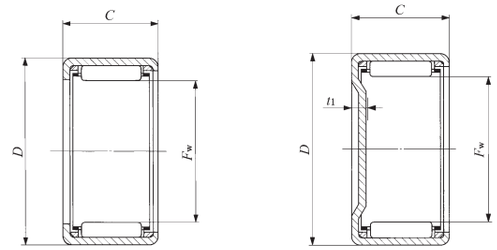
SHELL TYPE NEEDLE ROLLER BEARINGS





TA···Z TLA···Z TAM TLAM

Shaft dia. 50 — 62mm

Shaft dia. mm	Identification number									
	Standard	Mass (Ref.) g	Closed end	Mass (Ref.) g	Standard	Mass (Ref.) g	Closed end	Mass (Ref.) g	Grease retained	Mass (Ref.) g
50	—	—	—	—	**TLA 5020 Z**	69	**TLAM 5020**	84.5	—	—
	—	—	—	—	**TLA 5025 Z**	86	**TLAM 5025**	107	—	—
	TA 5012 Z	62.5	**TAM 5012**	78	—	—	—	—	—	—
	TA 5015 Z	78	**TAM 5015**	98.5	—	—	—	—	—	—
	TA 5020 Z	107	**TAM 5020**	123	—	—	—	—	—	—
	TA 5025 Z	134	**TAM 5025**	150	—	—	—	—	—	—
	TA 5030 Z	161	**TAM 5030**	178	—	—	—	—	—	—
	TA 5040 Z	210	**TAM 5040**	230	—	—	—	—	—	—
	TAW 5045 Z	230	**TAMW 5045**	245	—	—	—	—	—	—
55	—	—	—	—	**TLA 5520 Z**	75	**TLAM 5520**	98.5	—	—
	—	—	—	—	**TLA 5525 Z**	98.5	**TLAM 5525**	118	—	—
	TA 5520 Z	116	**TAM 5520**	136	—	—	—	—	—	—
	TA 5525 Z	145	**TAM 5525**	165	—	—	—	—	—	—
	TA 5530 Z	175	**TAM 5530**	195	—	—	—	—	—	—
	TA 5540 Z	230	**TAM 5540**	250	—	—	—	—	—	—
	TAW 5545 Z	250	**TAMW 5545**	270	—	—	—	—	—	—
	TAW 5550 Z	280	**TAMW 5550**	300	—	—	—	—	—	—
60	**TA 6025 Z**	158	**TAM 6025**	182	—	—	—	—	—	—
	TA 6030 Z	191	**TAM 6030**	215	—	—	—	—	—	—
	TA 6040 Z	250	**TAM 6040**	275	—	—	—	—	—	—
	TAW 6045 Z	270	**TAMW 6045**	295	—	—	—	—	—	—
	TAW 6050 Z	305	**TAMW 6050**	330	—	—	—	—	—	—
62	**TA 6212 Z**	78	**TAM 6212**	107	—	—	—	—	—	—

Boundary dimensions mm				Standard mounting dimensions mm						Basic dynamic load rating	Basic static load rating	Allowable rotational speed(1)	Assembled inner ring
				Shaft dia. h6		Housing bore dia. J7		Housing bore dia. N7					
F_w	D	C	t_1 Max.	Max.	Min.	Max.	Min.	Max.	Min.	C N	C_0 N	rpm	
50	58	20	2.8	50.000	49.984	—	—	57.991	57.961	28 800	64 100	6 500	**IRT 4520**
50	58	25	2.8							36 900	88 400	6 500	**IRT 4525**
50	62	12	3.4							17 700	24 000	6 500	**IRT 4512**
50	62	15	3.4							25 800	39 000	6 500	**IRT 4515**
50	62	20	3.4							38 000	64 000	6 500	**IRT 4520**
50	62	25	3.4	50.000	49.984	62.018	61.988	—	—	49 100	89 000	6 500	**IRT 4525**
50	62	30	3.4							59 500	114 000	6 500	**IRT 4530**
50	62	40	3.4							76 500	157 000	6 500	**IRT 4540**
50	62	45	3.4							76 700	158 000	6 500	**IRT 4545**
55	63	20	2.8	55.000	54.981	—	—	62.991	62.961	29 800	69 400	5 500	**IRT 5020-1**
55	63	25	2.8							38 300	95 700	5 500	**IRT 5025-1**
55	67	20	3.4							39 600	69 700	5 500	**IRT 5020-1**
55	67	25	3.4							51 200	97 000	5 500	**IRT 5025-1**
55	67	30	3.4							62 000	124 000	5 500	**IRT 5030-1**
55	67	40	3.4	55.000	54.981	67.018	66.988	—	—	80 000	172 000	5 500	**IRT 5040-1**
55	67	45	3.4							79 900	172 000	5 500	**IRT 5045-1**
55	67	50	3.4							91 500	205 000	5 500	**IRT 5050-1**
60	72	25	3.4							54 700	108 000	5 000	**IRT 5025**
60	72	30	3.4							66 300	139 000	5 000	**IRT 5030**
60	72	40	3.4	60.000	59.981	72.018	71.988	—	—	85 700	193 000	5 000	**IRT 5040**
60	72	45	3.4							85 400	193 000	5 000	**IRT 5045**
60	72	50	3.4							97 800	229 000	5 000	**IRT 5050**
62	74	12	3.4	62.000	61.981	74.018	73.988	—	—	20 100	30 300	4 500	**IRT 5212**

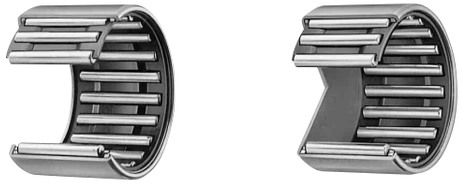
Note(1) Allowable rotational speed applies to oil lubrication. For grease lubrication, a maximum of 60% of this value is allowable.

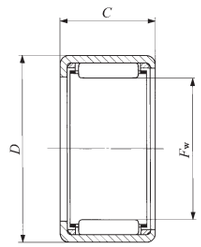
Remarks1. "W" in the identification number indicates that rolling elements are arranged in double rows.

2. Shell Type Grease Retained Full Complement Needle Roller Bearings are provided with prepacked grease. Standard type and closed end type bearings are not provided with prepacked grease, so perform proper lubrication when using these types of bearings.

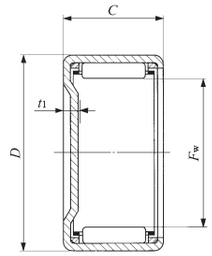
1N=0.102kgf=0.2248lbs.
1mm=0.03937inch

SHELL TYPE NEEDLE ROLLER BEARINGS





TA···Z



TAM

Shaft dia. 65 — 70mm

Shaft dia. mm	Identification number: Standard	Mass (Ref.) g	Closed end	Mass (Ref.) g	Standard	Mass (Ref.) g	Closed end	Mass (Ref.) g	Grease retained	Mass (Ref.) g
65	**TA 6525 Z**	169	**TAM 6525**	197	—	—	—	—	—	—
	TA 6530 Z	205	**TAM 6530**	230	—	—	—	—	—	—
	TAW 6545 Z	290	**TAMW 6545**	315	—	—	—	—	—	—
	TAW 6550 Z	330	**TAMW 6550**	355	—	—	—	—	—	—
70	**TA 7025 Z**	181	**TAM 7025**	215	—	—	—	—	—	—
	TA 7030 Z	220	**TAM 7030**	250	—	—	—	—	—	—
	TA 7040 Z	290	**TAM 7040**	320	—	—	—	—	—	—
	TAW 7050 Z	350	**TAMW 7050**	380	—	—	—	—	—	—

Boundary dimensions mm F_w	D	C	t_1 Max.	Standard mounting dimensions mm: Shaft dia. h6 Max.	Shaft dia. h6 Min.	Housing bore dia. J7 Max.	Housing bore dia. J7 Min.	Housing bore dia. N7 Max.	Housing bore dia. N7 Min.	Basic dynamic load rating C N	Basic static load rating C_0 N	Allowable rotational speed(1) rpm	Assembled inner ring
65	77	25	3.4	65.000	64.981	77.018	76.988	—	—	56 500	116 000	4 000	**IRT 5525**
65	77	30	3.4							68 500	149 000	4 000	**IRT 5530**
65	77	45	3.4							88 300	207 000	4 000	**IRT 5545**
65	77	50	3.4							101 000	246 000	4 000	**IRT 5550**
70	82	25	3.4	70.000	69.981	82.022	81.987	—	—	58 500	124 000	3 500	**IRT 6025**
70	82	30	3.4							70 900	159 000	3 500	**IRT 6030**
70	82	40	3.4							92 000	222 000	3 500	**IRT 6040**
70	82	50	3.4							105 000	262 000	3 500	**IRT 6050**

Note(1) Allowable rotational speed applies to oil lubrication. For grease lubrication, a maximum of 60% of this value is allowable.

Remarks1. "W" in the identification number indicates that rolling elements are arranged in double rows.

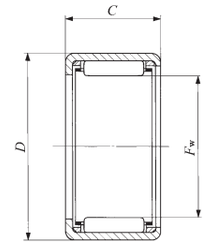
2. Shell Type Grease Retained Full Complement Needle Roller Bearings are provided with prepacked grease. Standard type and closed end type bearings are not provided with prepacked grease, so perform proper lubrication when using these types of bearings.

1N=0.102kgf=0.2248lbs.
1mm=0.03937inch

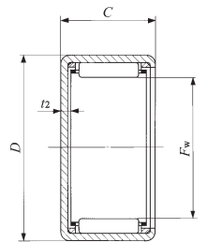
SHELL TYPE NEEDLE ROLLER BEARINGS

Inch Series

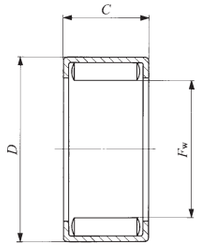




BA···Z BHA···Z



BAM BHAM



YB

Shaft dia. 3.969 — 9.525mm

Shaft dia. mm (inch)	Identification number									
	Standard	Mass (Ref.) g	Closed end	Mass (Ref.) g	Standard	Mass (Ref.) g	Closed end	Mass (Ref.) g	Grease retained	Mass (Ref.) g
3.969 (5/32)	—	—	—	—	—	—	—	—	**YB 2.5 2.5**	0.64
	—	—	—	—	—	—	—	—	**YB 2.5 4**	0.96
4.762 (3/16)	—	—	—	—	—	—	—	—	**YB 34**	1.6
6.350 (1/4)	**BA 44**	2.1	—	—	—	—	—	—	—	—
	BA 45 Z	2.5	**BAM 45**	2.7	—	—	—	—	—	—
	BA 47 Z	3.5	**BAM 47**	3.7	—	—	—	—	—	—
	—	—	—	—	—	—	—	—	**YB 45**	3.2
	—	—	—	—	—	—	—	—	**YB 47**	4.6
7.938 (5/16)	**BA 55 Z**	3	**BAM 55**	3.3	—	—	—	—	—	—
	BA 56 Z	3.6	**BAM 56**	3.9	—	—	—	—	—	—
	BA 57 Z	4.3	**BAM 57**	4.6	—	—	—	—	—	—
	BA 59 Z	5.4	**BAM 59**	5.7	—	—	—	—	—	—
	—	—	—	—	—	—	—	—	**YB 55**	3.8
	—	—	—	—	**BHA 57 Z**	6.3	**BHAM 57**	6.6	—	—
9.525 (3/8)	**BA 65 Z**	3.5	**BAM 65**	3.9	—	—	—	—	—	—
	BA 66 Z	4.2	**BAM 66**	4.6	—	—	—	—	—	—
	BA 68 Z	5.7	**BAM 68**	6.1	—	—	—	—	—	—
	BA 69 Z	6.3	**BAM 69**	6.7	—	—	—	—	—	—
	BA 610 Z	7	**BAM 610**	7.4	—	—	—	—	—	—
	—	—	—	—	—	—	—	—	**YB 64**	3.4
	—	—	—	—	—	—	—	—	**YB 66**	5.3
	—	—	—	—	—	—	—	—	**YB 68**	7.2
	—	—	—	—	—	—	—	—	**YB 610**	9.1
	—	—	—	—	**BHA 68 Z**	8.2	**BHAM 68**	8.6	—	—

Boundary dimensions mm(inch)				Standard mounting dimensions mm				Basic dynamic load rating	Basic static load rating	Allowable rotational speed(1)	Assembled inner ring
				Shaft dia. h6		Housing bore dia. J7					
F_w	D	C	t_2 Max.	Max.	Min.	Max.	Min.	C N	C_0 N	rpm	
3.969 (5/32)	7.144 (9/32)	3.96(.156)	—	3.969	3.961	7.152	7.137	1 350	1 220	40 000	—
3.969 (5/32)	7.144 (9/32)	6.35(.250)	—					2 320	2 440	40 000	—
4.762 (3/16)	8.731 (11/32)	6.35(.250)	—	4.762	4.754	8.739	8.724	2 770	2 700	30 000	—
6.350 (1/4)	11.112 (7/16)	6.35(.250)	1	6.350	6.341	11.122	11.104	1 770	1 390	55 000	—
6.350 (1/4)	11.112 (7/16)	7.92(.312)	1					1 510	1 120	55 000	—
6.350 (1/4)	11.112 (7/16)	11.13(.438)	1					2 650	2 310	55 000	—
6.350 (1/4)	11.112 (7/16)	7.92(.312)	—					4 450	4 870	25 000	—
6.350 (1/4)	11.112 (7/16)	11.13(.438)	—					6 320	7 650	25 000	—
7.938 (5/16)	12.700 (1/2)	7.92(.312)	1	7.938	7.929	12.710	12.692	1 880	1 560	45 000	—
7.938 (5/16)	12.700 (1/2)	9.52(.375)	1					2 620	2 390	45 000	—
7.938 (5/16)	12.700 (1/2)	11.13(.438)	1					3 310	3 220	45 000	—
7.938 (5/16)	12.700 (1/2)	14.27(.562)	1					4 190	4 360	45 000	—
7.938 (5/16)	12.700 (1/2)	7.92(.312)	—					5 110	6 090	20 000	—
7.938 (5/16)	14.288 (9/16)	11.13(.438)	1.3	7.938	7.929	14.298	14.280	4 150	3 730	45 000	—
9.525 (3/8)	14.288 (9/16)	7.92(.312)	1	9.525	9.516	14.298	14.280	2 220	2 010	40 000	—
9.525 (3/8)	14.288 (9/16)	9.52(.375)	1					3 090	3 080	40 000	—
9.525 (3/8)	14.288 (9/16)	12.70(.500)	1					4 190	4 560	40 000	—
9.525 (3/8)	14.288 (9/16)	14.27(.562)	1					4 940	5 630	40 000	—
9.525 (3/8)	14.288 (9/16)	15.88(.625)	1					5 660	6 700	40 000	—
9.525 (3/8)	14.288 (9/16)	6.35(.250)	—	9.525	9.516	14.298	14.280	4 470	5 360	16 000	—
9.525 (3/8)	14.288 (9/16)	9.52(.375)	—					6 920	9 410	16 000	—
9.525 (3/8)	14.288 (9/16)	12.70(.500)	—					9 210	13 600	16 000	—
9.525 (3/8)	14.288 (9/16)	15.88(.625)	—					11 300	17 800	16 000	—
9.525 (3/8)	15.875 (5/8)	12.70(.500)	1.3	9.525	9.516	15.885	15.867	4 880	4 740	40 000	—

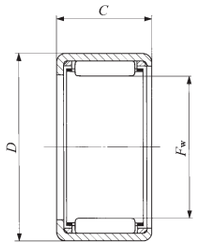
Note(1) Allowable rotational speed applies to oil lubrication. For grease lubrication, a maximum of 60% of this value is allowable.
Remark Shell Type Grease Retained Full Complement Needle Roller Bearings are provided with prepacked grease. Standard type and closed end type bearings are not provided with prepacked grease, so perform proper lubrication when using these types of bearings.

1N=0.102kgf=0.2248lbs.
1mm=0.03937inch

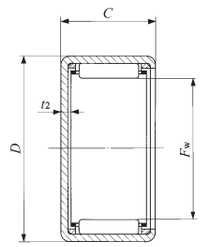
SHELL TYPE NEEDLE ROLLER BEARINGS

Inch Series

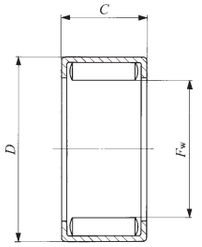




BA···Z BHA···Z



BAM BHAM



YB YBH

Shaft dia. 11.112 — 12.700mm

Shaft dia. mm (inch)	Identification number									
	Standard	Mass (Ref.) g	Closed end	Mass (Ref.) g	Standard	Mass (Ref.) g	Closed end	Mass (Ref.) g	Grease retained	Mass (Ref.) g
11.112 (7/16)	**BA 76 Z**	4.8	**BAM 76**	5.3	—	—	—	—	—	—
	BA 77 Z	5.6	**BAM 77**	6.2	—	—	—	—	—	—
	BA 78 Z	6.4	**BAM 78**	7	—	—	—	—	—	—
	BA 710 Z	7.9	**BAM 710**	8.5	—	—	—	—	—	—
	—	—	—	—	—	—	—	—	**YB 78**	8.2
	—	—	—	—	**BHA 78 Z**	9.3	**BHAM 78**	10	—	—
	—	—	—	—	—	—	—	—	**YBH 78**	10.5
12.700 (1/2)	**BA 85 Z**	4.4	**BAM 85**	5.2	—	—	—	—	—	—
	BA 86 Z	5.3	**BAM 86**	6.1	—	—	—	—	—	—
	BA 87 Z	6.3	**BAM 87**	7	—	—	—	—	—	—
	BA 88 Z	7.2	**BAM 88**	7.9	—	—	—	—	—	—
	BA 810 Z	8.9	**BAM 810**	9.6	—	—	—	—	—	—
	BA 812 Z	10.6	**BAM 812**	11.3	—	—	—	—	—	—
	—	—	—	—	—	—	—	—	**YB 84**	4.3
	—	—	—	—	—	—	—	—	**YB 86**	6.7
	—	—	—	—	—	—	—	—	**YB 87**	7.9
	—	—	—	—	—	—	—	—	**YB 88**	9.1
	—	—	—	—	—	—	—	—	**YB 810**	11.5
	—	—	—	—	—	—	—	—	**YB 812**	13.9
	—	—	—	—	**BHA 87 Z**	9.1	**BHAM 87**	9.9	—	—
	—	—	—	—	**BHA 88 Z**	10.4	**BHAM 88**	11.3	—	—
	—	—	—	—	**BHA 810 Z**	12.5	**BHAM 810**	13.3	—	—
	—	—	—	—	**BHA 812 Z**	15	**BHAM 812**	15.8	—	—
	—	—	—	—	—	—	—	—	**YBH 810**	16

Boundary dimensions mm(inch)				Standard mounting dimensions mm				Basic dynamic load rating	Basic static load rating	Allowable rotational speed(1)	Assembled inner ring
				Shaft dia. h6		Housing bore dia. J7					
F_w	D	C	t_2 Max.	Max.	Min.	Max.	Min.	C N	C_0 N	rpm	
11.112 (7/16)	15.875 (5/8)	9.52(.375)	1					3 290	3 470	35 000	—
11.112 (7/16)	15.875 (5/8)	11.13(.438)	1					4 150	4 680	35 000	—
11.112 (7/16)	15.875 (5/8)	12.70(.500)	1	11.112	11.101	15.885	15.867	4 460	5 130	35 000	—
11.112 (7/16)	15.875 (5/8)	15.88(.625)	1					6 020	7 550	35 000	—
11.112 (7/16)	15.875 (5/8)	12.70(.500)	—					10 100	15 900	14 000	—
11.112 (7/16)	17.462 (11/16)	12.70(.500)	1.3	11.112	11.101	17.472	17.454	5 680	5 970	35 000	—
11.112 (7/16)	17.462 (11/16)	12.70(.500)	—					12 500	15 800	14 000	—
12.700 (1/2)	17.462 (11/16)	7.92(.312)	1					2 490	2 510	30 000	—
12.700 (1/2)	17.462 (11/16)	9.52(.375)	1					3 470	3 850	30 000	—
12.700 (1/2)	17.462 (11/16)	11.13(.438)	1	12.700	12.689	17.472	17.454	4 380	5 190	30 000	—
12.700 (1/2)	17.462 (11/16)	12.70(.500)	1					4 710	5 700	30 000	**IRB 58**
12.700 (1/2)	17.462 (11/16)	15.88(.625)	1					6 350	8 380	30 000	—
12.700 (1/2)	17.462 (11/16)	19.05(.750)	1					7 840	11 000	30 000	—
12.700 (1/2)	17.462 (11/16)	6.35(.250)	—					5 260	7 150	12 000	—
12.700 (1/2)	17.462 (11/16)	9.52(.375)	—					8 150	12 600	12 000	—
12.700 (1/2)	17.462 (11/16)	11.13(.438)	—	12.700	12.689	17.472	17.454	9 530	15 300	12 000	—
12.700 (1/2)	17.462 (11/16)	12.70(.500)	—					10 800	18 100	12 000	**IRB 58**
12.700 (1/2)	17.462 (11/16)	15.88(.625)	—					13 400	23 700	12 000	—
12.700 (1/2)	17.462 (11/16)	19.05(.750)	—					15 800	29 300	12 000	—
12.700 (1/2)	19.050 (3/4)	11.13(.438)	1.3					5 670	6 120	30 000	—
12.700 (1/2)	19.050 (3/4)	12.70(.500)	1.3					6 040	6 650	30 000	**IRB 58**
12.700 (1/2)	19.050 (3/4)	15.88(.625)	1.3	12.700	12.689	19.062	19.041	8 830	10 900	30 000	—
12.700 (1/2)	19.050 (3/4)	19.05(.750)	1.3					11 100	14 500	30 000	—
12.700 (1/2)	19.050 (3/4)	15.88(.625)	—					16 300	23 500	12 000	—

Note(1) Allowable rotational speed applies to oil lubrication. For grease lubrication, a maximum of 60% of this value is allowable.
Remark Shell Type Grease Retained Full Complement Needle Roller Bearings are provided with prepacked grease. Standard type and closed end type bearings are not provided with prepacked grease, so perform proper lubrication when using these types of bearings.

1N=0.102kgf=0.2248lbs.
1mm=0.03937inch

SHELL TYPE NEEDLE ROLLER BEARINGS

Inch Series

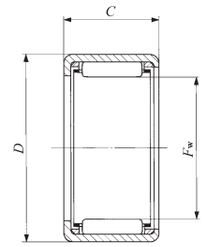


Shaft dia. 14.288 — 15.875mm

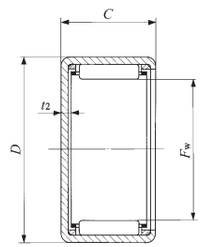
Shaft dia. mm (inch)	Identification number									
	Standard	Mass (Ref.) g	Closed end	Mass (Ref.) g	Standard	Mass (Ref.) g	Closed end	Mass (Ref.) g	Grease retained	Mass (Ref.) g
14.288 (9/16)	BA 95 Z	4.9	BAM 95	5.8	—	—	—	—	—	—
	BA 96 Z	5.9	BAM 96	6.8	—	—	—	—	—	—
	BA 97 Z	6.9	BAM 97	7.8	—	—	—	—	—	—
	BA 98 Z	7.9	BAM 98	8.9	—	—	—	—	—	—
	BA 910 Z	9.9	BAM 910	10.8	—	—	—	—	—	—
	BA 912 Z	11.7	BAM 912	12.6	—	—	—	—	—	—
	—	—	—	—	—	—	—	—	YB 98	10.1
	—	—	—	—	—	—	—	—	YB 910	12.7
	—	—	—	—	—	—	—	—	YB 912	15.4
	—	—	—	—	BHA 98 Z	11.4	BHAM 98	12.5	—	—
	—	—	—	—	BHA 910 Z	13.6	BHAM 910	14.7	—	—
	—	—	—	—	BHA 912 Z	16.3	BHAM 912	17.4	—	—
15.875 (5/8)	BA 105 Z	5.3	BAM 105	6.5	—	—	—	—	—	—
	BA 107 Z	7.6	BAM 107	8.7	—	—	—	—	—	—
	BA 108 Z	8.7	BAM 108	9.9	—	—	—	—	—	—
	BA 1010 Z	10.8	BAM 1010	12	—	—	—	—	—	—
	BA 1012 Z	12.9	BAM 1012	14	—	—	—	—	—	—
	BA 1014 Z	15.1	BAM 1014	16.2	—	—	—	—	—	—
	BA 1016 Z	17.3	BAM 1016	18.4	—	—	—	—	—	—
	—	—	—	—	—	—	—	—	YB 105	6.7
	—	—	—	—	—	—	—	—	YB 108	11
	—	—	—	—	—	—	—	—	YB 1012	16.9
	—	—	—	—	BHA 108 Z	12.6	BHAM 108	13.9	—	—
	—	—	—	—	BHA 1010 Z	14.9	BHAM 1010	16.2	—	—
	—	—	—	—	BHA 1012 Z	18	BHAM 1012	19.3	—	—
	—	—	—	—	BHA 1016 Z	24	BHAM 1016	25	—	—
	—	—	—	—	—	—	—	—	YBH 108	15.3

Note(1) Allowable rotational speed applies to oil lubrication. For grease lubrication, a maximum of 60% of this value is allowable.

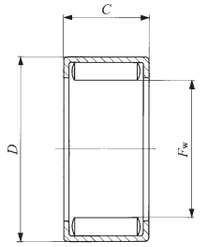
Remark Shell Type Grease Retained Full Complement Needle Roller Bearings are provided with prepacked grease. Standard type and closed end type bearings are not provided with prepacked grease, so perform proper lubrication when using these types of bearings.



BA···Z BHA···Z



BAM BHAM



YB YBH

Boundary dimensions mm(inch)				Standard mounting dimensions mm				Basic dynamic load rating	Basic static load rating	Allowable rotational speed(1)	Assembled inner ring
				Shaft dia. h6		Housing bore dia. J7					
F_w	D	C	t_2 Max.	Max.	Min.	Max.	Min.	C N	C_0 N	rpm	
14.288 (9/16)	19.050 (3/4)	7.92(.312)	1.3					2 760	2 970	30 000	—
14.288 (9/16)	19.050 (3/4)	9.52(.375)	1.3					3 850	4 560	30 000	—
14.288 (9/16)	19.050 (3/4)	11.13(.438)	1.3					4 860	6 140	30 000	—
14.288 (9/16)	19.050 (3/4)	12.70(.500)	1.3					5 220	6 740	30 000	IRB 68
14.288 (9/16)	19.050 (3/4)	15.88(.625)	1.3	14.288	14.277	19.062	19.041	7 050	9 910	30 000	—
14.288 (9/16)	19.050 (3/4)	19.05(.750)	1.3					8 690	13 000	30 000	IRB 612
14.288 (9/16)	19.050 (3/4)	12.70(.500)	—					11 600	20 400	11 000	IRB 68
14.288 (9/16)	19.050 (3/4)	15.88(.625)	—					14 300	26 700	11 000	—
14.288 (9/16)	19.050 (3/4)	19.05(.750)	—					16 800	33 000	11 000	IRB 612
14.288 (9/16)	20.638 (13/16)	12.70(.500)	1.3					6 380	7 330	30 000	IRB 68
14.288 (9/16)	20.638 (13/16)	15.88(.625)	1.3	14.288	14.277	20.650	20.629	9 280	11 900	30 000	—
14.288 (9/16)	20.638 (13/16)	19.05(.750)	1.3					11 600	15 900	30 000	IRB 612
15.875 (5/8)	20.638 (13/16)	7.92(.312)	1.3					2 870	3 220	25 000	—
15.875 (5/8)	20.638 (13/16)	11.13(.438)	1.3					5 040	6 660	25 000	—
15.875 (5/8)	20.638 (13/16)	12.70(.500)	1.3					5 420	7 310	25 000	IRB 68-1
15.875 (5/8)	20.638 (13/16)	15.88(.625)	1.3					7 320	10 700	25 000	—
15.875 (5/8)	20.638 (13/16)	19.05(.750)	1.3	15.875	15.864	20.650	20.629	9 020	14 100	25 000	IRB 612-1
15.875 (5/8)	20.638 (13/16)	22.22(.875)	1.3					10 700	17 500	25 000	IRB 714
15.875 (5/8)	20.638 (13/16)	25.40(1.000)	1.3					12 300	20 800	25 000	IRB 716
15.875 (5/8)	20.638 (13/16)	7.92(.312)	—					7 580	12 200	9 500	—
15.875 (5/8)	20.638 (13/16)	12.70(.500)	—					12 300	22 700	9 500	IRB 68-1
15.875 (5/8)	20.638 (13/16)	19.05(.750)	—					17 800	36 600	9 500	IRB 612-1
15.875 (5/8)	22.225 (7/8)	12.70(.500)	1.3					6 680	8 020	25 000	IRB 68-1
15.875 (5/8)	22.225 (7/8)	15.88(.625)	1.3					10 200	13 800	25 000	—
15.875 (5/8)	22.225 (7/8)	19.05(.750)	1.3	15.875	15.864	22.237	22.216	12 700	18 500	25 000	IRB 612-1
15.875 (5/8)	22.225 (7/8)	25.40(1.000)	1.3					17 400	27 600	25 000	IRB 716
15.875 (5/8)	22.225 (7/8)	12.70(.500)	—					15 000	22 400	9 500	IRB 68-1

1N=0.102kgf=0.2248lbs.
1mm=0.03937inch

SHELL TYPE NEEDLE ROLLER BEARINGS

Inch Series

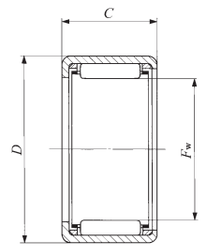


Shaft dia. 17.462 — 19.050mm

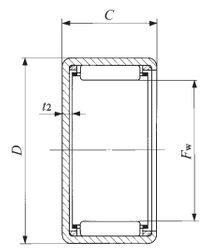
Shaft dia. mm (inch)	Identification number									
	Standard	Mass (Ref.) g	Closed end	Mass (Ref.) g	Standard	Mass (Ref.) g	Closed end	Mass (Ref.) g	Grease retained	Mass (Ref.) g
17.462 (11/16)	**BA 116 Z**	7	**BAM 116**	8.4	—	—	—	—	—	—
	BA 118 Z	9.5	**BAM 118**	10.8	—	—	—	—	—	—
	BA 1110 Z	11.8	**BAM 1110**	13.2	—	—	—	—	—	—
	BA 1112 Z	14	**BAM 1112**	15.4	—	—	—	—	—	—
	—	—	—	—	—	—	—	—	**YB 1112**	18.3
	—	—	—	—	**BHA 117 Z**	11.9	**BHAM 117**	13.5	—	—
	—	—	—	—	**BHA 118 Z**	13.7	**BHAM 118**	15.3	—	—
	—	—	—	—	**BHA 1110 Z**	16	**BHAM 1110**	17.6	—	—
	—	—	—	—	**BHA 1112 Z**	19.3	**BHAM 1112**	21	—	—
19.050 (3/4)	**BA 126 Z**	10	**BAM 126**	11.7	—	—	—	—	—	—
	BA 128 Z	13.5	**BAM 128**	15.2	—	—	—	—	—	—
	BA 1210 Z	17	**BAM 1210**	18.6	—	—	—	—	—	—
	BA 1212 Z	20.5	**BAM 1212**	22	—	—	—	—	—	—
	BA 1214 Z	23.5	**BAM 1214**	25	—	—	—	—	—	—
	BA 1216 Z	27	**BAM 1216**	28.5	—	—	—	—	—	—
	—	—	—	—	—	—	—	—	**YB 124**	8.5
	—	—	—	—	—	—	—	—	**YB 128**	17.8
	—	—	—	—	—	—	—	—	**YB 1210**	22.5
	—	—	—	—	—	—	—	—	**YB 1212**	27
	—	—	—	—	**BHA 1212 Z**	26.5	**BHAM 1212**	28.5	—	—

Note(1) Allowable rotational speed applies to oil lubrication. For grease lubrication, a maximum of 60% of this value is allowable.

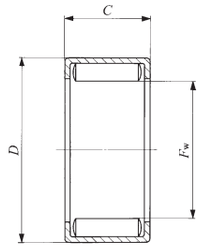
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BA···Z BHA···Z



BAM BHAM



YB

Boundary dimensions mm(inch)				Standard mounting dimensions mm				Basic dynamic load rating	Basic static load rating	Allowable rotational speed(1)	Assembled inner ring
				Shaft dia. h6		Housing bore dia. J7					
F_w	D	C	t_2 Max.	Max.	Min.	Max.	Min.	C N	C_0 N	rpm	
17.462 (11/16)	22.225 (7/8)	9.52 (.375)	1.3					4 530	5 980	25 000	**IRB 86**
17.462 (11/16)	22.225 (7/8)	12.70 (.500)	1.3					6 140	8 850	25 000	**IRB 88**
17.462 (11/16)	22.225 (7/8)	15.88 (.625)	1.3	17.462	17.451	22.237	22.216	8 280	13 000	25 000	—
17.462 (11/16)	22.225 (7/8)	19.05 (.750)	1.3					10 200	17 000	25 000	**IRB 812**
17.462 (11/16)	22.225 (7/8)	19.05 (.750)	—					18 700	40 300	8 500	**IRB 812**
17.462 (11/16)	23.812 (15/16)	11.13 (.438)	1.3					6 860	8 530	25 000	—
17.462 (11/16)	23.812 (15/16)	12.70 (.500)	1.3					7 320	9 270	25 000	**IRB 88**
17.462 (11/16)	23.812 (15/16)	15.88 (.625)	1.3	17.462	17.451	23.824	23.803	10 500	14 900	25 000	—
17.462 (11/16)	23.812 (15/16)	19.05 (.750)	1.3					13 200	19 900	25 000	**IRB 812**
19.050 (3/4)	25.400 (1)	9.52 (.375)	1.3					5 040	5 850	20 000	—
19.050 (3/4)	25.400 (1)	12.70 (.500)	1.3					6 910	8 780	20 000	**IRB 88-1**
19.050 (3/4)	25.400 (1)	15.88 (.625)	1.3					9 500	13 200	20 000	**IRB 810-1**
19.050 (3/4)	25.400 (1)	19.05 (.750)	1.3	19.050	19.037	25.412	25.391	11 900	17 700	20 000	**IRB 812-1**
19.050 (3/4)	25.400 (1)	22.22 (.875)	1.3					14 200	22 200	20 000	**IRB 814-1**
19.050 (3/4)	25.400 (1)	25.40 (1.000)	1.3					16 300	26 500	20 000	**IRB 816-1**
19.050 (3/4)	25.400 (1)	6.35 (.250)	—					7 820	10 200	8 000	—
19.050 (3/4)	25.400 (1)	12.70 (.500)	—					16 600	26 900	8 000	**IRB 88-1**
19.050 (3/4)	25.400 (1)	15.88 (.625)	—	19.050	19.037	25.412	25.391	20 500	35 300	8 000	**IRB 810-1**
19.050 (3/4)	25.400 (1)	19.05 (.750)	—					24 100	43 400	8 000	**IRB 812-1**
19.050 (3/4)	26.988 (1 1/16)	19.05 (.750)	1.3	19.050	19.037	27.000	26.979	16 600	22 600	20 000	**IRB 812-1**

1N=0.102kgf=0.2248lbs.
1mm=0.03937inch

SHELL TYPE NEEDLE ROLLER BEARINGS

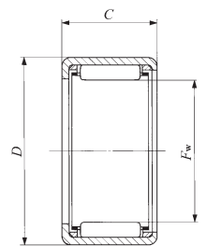
Inch Series



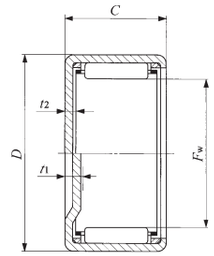
Shaft dia. 20.638 — 22.225mm

Shaft dia. mm (inch)	Identification number									
	Standard	Mass (Ref.) g	Closed end	Mass (Ref.) g	Standard	Mass (Ref.) g	Closed end	Mass (Ref.) g	Grease retained	Mass (Ref.) g
20.638 (13/16)	**BA 136 Z**	10.7	**BAM 136**	12.6	—	—	—	—	—	—
	BA 138 Z	14.5	**BAM 138**	16.4	—	—	—	—	—	—
	BA 1310 Z	18.2	**BAM 1310**	20	—	—	—	—	—	—
	BA 1312 Z	22	**BAM 1312**	23.5	—	—	—	—	—	—
	BA 1314 Z	25	**BAM 1314**	27	—	—	—	—	—	—
	BA 1316 Z	28.5	**BAM 1316**	30.5	—	—	—	—	—	—
	BA 1320 Z	35.5	**BAM 1320**	37.5	—	—	—	—	—	—
	—	—	—	—	—	—	—	—	**YB 136**	14.1
	—	—	—	—	—	—	—	—	**YB 138**	19.1
	—	—	—	—	**BHA 138 Z**	20	**BHAM 138**	22.5	—	—
	—	—	—	—	**BHA 1310 Z**	23.5	**BHAM 1310**	25.5	—	—
	—	—	—	—	**BHA 1312 Z**	28.5	**BHAM 1312**	30.5	—	—
	—	—	—	—	—	—	—	—	**YBH 1310**	30.5
	—	—	—	—	—	—	—	—	**YBH 1312**	37
22.225 (7/8)	**BA 146 Z**	11.5	**BAM 146**	13.8	—	—	—	—	—	—
	BA 148 Z	15.6	**BAM 148**	17.8	—	—	—	—	—	—
	BA 1412 Z	23.5	**BAM 1412**	26	—	—	—	—	—	—
	BA 1414 Z	27	**BAM 1414**	29.5	—	—	—	—	—	—
	BA 1416 Z	31	**BAM 1416**	33.5	—	—	—	—	—	—
	BA 1418 Z	34.5	**BAM 1418**	37	—	—	—	—	—	—
	BA 1422 Z	42.5	**BAM 1422**	44.5	—	—	—	—	—	—
	—	—	—	—	—	—	—	—	**YB 148**	20.5
	—	—	—	—	—	—	—	—	**YB 1412**	31
	—	—	—	—	—	—	—	—	**YB 1416**	41.5
	—	—	—	—	**BHA 1410 Z**	25	**BHAM 1410**	27.5	—	—
	—	—	—	—	**BHA 1412 Z**	30	**BHAM 1412**	32.5	—	—
	—	—	—	—	**BHA 1416 Z**	39.5	**BHAM 1416**	42	—	—
	—	—	—	—	—	—	—	—	**YBH 1412**	39

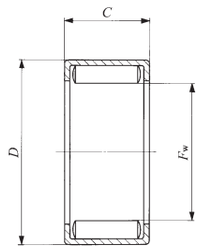
Note(1) Allowable rotational speed applies to oil lubrication. For grease lubrication, a maximum of 60% of this value is allowable.
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BA···Z BHA···Z



BAM BHAM
t_1 ($F_w \geqq 22.225$)
t_2 ($F_w \leqq 20.638$)



YB YBH

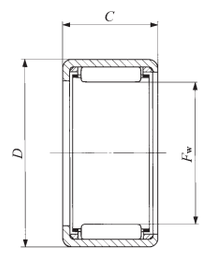
Boundary dimensions mm(inch)				Standard mounting dimensions mm				Basic dynamic load rating	Basic static load rating	Allowable rotational speed(1)	Assembled inner ring
				Shaft dia. h6		Housing bore dia. J7					
F_w	D	C	t_1 t_2 Max.	Max.	Min.	Max.	Min.	C N	C_0 N	rpm	
20.638 (13/16)	26.988 (1 1/16)	9.52(.375)	1.3					5 230	6 300	19 000	—
20.638 (13/16)	26.988 (1 1/16)	12.70(.500)	1.3					7 170	9 450	19 000	**IRB 98**
20.638 (13/16)	26.988 (1 1/16)	15.88(.625)	1.3					9 870	14 200	19 000	**IRB 910**
20.638 (13/16)	26.988 (1 1/16)	19.05(.750)	1.3					12 400	19 000	19 000	**IRB 912**
20.638 (13/16)	26.988 (1 1/16)	22.22(.875)	1.3	20.638	20.625	27.000	26.979	14 700	23 800	19 000	**IRB 914**
20.638 (13/16)	26.988 (1 1/16)	25.40(1.000)	1.3					16 900	28 500	19 000	**IRB 916**
20.638 (13/16)	26.988 (1 1/16)	31.75(1.250)	1.3					21 200	38 100	19 000	**IRB 920**
20.638 (13/16)	26.988 (1 1/16)	9.52(.375)	—					13 000	20 100	7 500	—
20.638 (13/16)	26.988 (1 1/16)	12.70(.500)	—					17 400	29 200	7 500	**IRB 98**
20.638 (13/16)	28.575 (1 1/8)	12.70(.500)	1.3					9 500	11 200	19 000	**IRB 98**
20.638 (13/16)	28.575 (1 1/8)	15.88(.625)	1.3					13 800	18 200	19 000	**IRB 910**
20.638 (13/16)	28.575 (1 1/8)	19.05(.750)	1.3	20.638	20.625	28.587	28.566	17 300	24 400	19 000	**IRB 912**
20.638 (13/16)	28.575 (1 1/8)	15.88(.625)	—					22 900	36 300	7 500	**IRB 910**
20.638 (13/16)	28.575 (1 1/8)	19.05(.750)	—					27 200	45 300	7 500	**IRB 912**
22.225 (7/8)	28.575 (1 1/8)	9.52(.375)	2.8					5 430	6 740	18 000	**IRB 106**
22.225 (7/8)	28.575 (1 1/8)	12.70(.500)	2.8					7 440	10 100	18 000	**IRB 108**
22.225 (7/8)	28.575 (1 1/8)	19.05(.750)	2.8					12 800	20 400	18 000	**IRB 1012**
22.225 (7/8)	28.575 (1 1/8)	22.22(.875)	2.8					15 300	25 500	18 000	**IRB 1014**
22.225 (7/8)	28.575 (1 1/8)	25.40(1.000)	2.8	22.225	22.212	28.587	28.566	17 600	30 500	18 000	**IRB 1016**
22.225 (7/8)	28.575 (1 1/8)	28.58(1.125)	2.8					19 800	35 600	18 000	—
22.225 (7/8)	28.575 (1 1/8)	34.92(1.375)	2.8					24 100	45 700	18 000	**IRB 1022**
22.225 (7/8)	28.575 (1 1/8)	12.70(.500)	—					18 100	31 400	7 000	**IRB 108**
22.225 (7/8)	28.575 (1 1/8)	19.05(.750)	—					26 300	50 700	7 000	**IRB 1012**
22.225 (7/8)	28.575 (1 1/8)	25.40(1.000)	—					33 800	70 200	7 000	**IRB 1016**
22.225 (7/8)	30.162 (1 3/16)	15.88(.625)	3.4					14 300	19 500	18 000	—
22.225 (7/8)	30.162 (1 3/16)	19.05(.750)	3.4	22.225	22.212	30.176	30.151	18 000	26 100	18 000	**IRB 1012**
22.225 (7/8)	30.162 (1 3/16)	25.40(1.000)	3.4					23 600	36 900	18 000	**IRB 1016**
22.225 (7/8)	30.162 (1 3/16)	19.05(.750)	—					28 200	49 000	7 000	**IRB 1012**

1N=0.102kgf=0.2248lbs.
1mm=0.03937inch

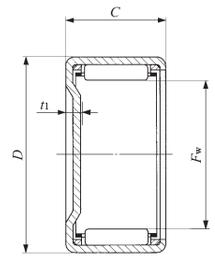
SHELL TYPE NEEDLE ROLLER BEARINGS

Inch Series

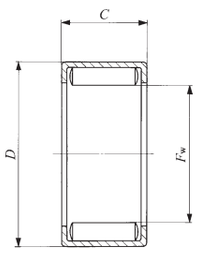




BA···Z BHA···Z



BAM BHAM



YB YBH

Shaft dia. 23.812 — 26.988mm

Shaft dia. mm (inch)	Identification number									
	Standard	Mass (Ref.) g	Closed end	Mass (Ref.) g	Standard	Mass (Ref.) g	Closed end	Mass (Ref.) g	Grease retained	Mass (Ref.) g
23.812 (15/16)	**BA 158 Z**	16.5	**BAM 158**	19	—	—	—	—	—	—
	BA 1510 Z	20.5	**BAM 1510**	23	—	—	—	—	—	—
	BA 1516 Z	33	**BAM 1516**	35.5	—	—	—	—	—	—
25.400 (1)	**BA 166 Z**	13.1	**BAM 166**	16	—	—	—	—	—	—
	BA 167 Z	15.4	**BAM 167**	18.3	—	—	—	—	—	—
	BA 168 Z	17.7	**BAM 168**	20.5	—	—	—	—	—	—
	BA 1610 Z	22	**BAM 1610**	25	—	—	—	—	—	—
	BA 1612 Z	26.5	**BAM 1612**	29.5	—	—	—	—	—	—
	BA 1614 Z	31	**BAM 1614**	33.5	—	—	—	—	—	—
	BA 1616 Z	35.5	**BAM 1616**	38	—	—	—	—	—	—
	BA 1620 Z	44	**BAM 1620**	46.5	—	—	—	—	—	—
	—	—	—	—	—	—	—	—	**YB 168**	23
	—	—	—	—	—	—	—	—	**YB 1612**	34.5
	—	—	—	—	—	—	—	—	**YB 1616**	46.5
	—	—	—	—	**BHA 168 Z**	24	**BHAM 168**	27	—	—
	—	—	—	—	**BHA 1610 Z**	28	**BHAM 1610**	31	—	—
	—	—	—	—	**BHA 1612 Z**	33.5	**BHAM 1612**	37	—	—
	—	—	—	—	**BHA 1614 Z**	39.5	**BHAM 1614**	42.5	—	—
	—	—	—	—	**BHA 1616 Z**	45	**BHAM 1616**	48	—	—
	—	—	—	—	**BHA 1620 Z**	56.5	**BHAM 1620**	59.5	—	—
	—	—	—	—	**BHA 1624 Z**	67.5	**BHAM 1624**	71	—	—
	—	—	—	—	—	—	—	—	**YBH 168**	29
	—	—	—	—	—	—	—	—	**YBH 1612**	44.5
	—	—	—	—	—	—	—	—	**YBH 1616**	59.5
26.988 (1 1/16)	**BA 1710 Z**	23.5	**BAM 1710**	26.5	—	—	—	—	—	—
	BA 1716 Z	37	**BAM 1716**	40.5	—	—	—	—	—	—

Boundary dimensions mm(inch)				Standard mounting dimensions mm				Basic dynamic load rating	Basic static load rating	Allowable rotational speed(1)	Assembled inner ring
				Shaft dia. h6		Housing bore dia. J7					
F_w	D	C	t_1 Max.	Max.	Min.	Max.	Min.	C N	C_0 N	rpm	
23.812 (15/16)	30.162 (1 3/16)	12.70 (.500)	2.8					8 000	11 400	16 000	—
23.812 (15/16)	30.162 (1 3/16)	15.88 (.625)	2.8	23.812	23.799	30.176	30.151	11 000	17 100	16 000	**IRB 1110**
23.812 (15/16)	30.162 (1 3/16)	25.40(1.000)	2.8					18 900	34 300	16 000	**IRB 1116**
25.400 (1)	31.750 (1 1/4)	9.52(.375)	2.8					6 010	8 020	15 000	—
25.400 (1)	31.750 (1 1/4)	11.13(.438)	2.8					7 720	11 100	15 000	—
25.400 (1)	31.750 (1 1/4)	12.70(.500)	2.8					8 240	12 000	15 000	**IRB 128**
25.400 (1)	31.750 (1 1/4)	15.88(.625)	2.8					11 300	18 100	15 000	—
25.400 (1)	31.750 (1 1/4)	19.05(.750)	2.8					14 200	24 300	15 000	**IRB 1212**
25.400 (1)	31.750 (1 1/4)	22.22(.875)	2.8	25.400	25.387	31.764	31.739	16 900	30 400	15 000	**IRB 1214**
25.400 (1)	31.750 (1 1/4)	25.40(1.000)	2.8					19 400	36 300	15 000	**IRB 1216**
25.400 (1)	31.750 (1 1/4)	31.75(1.250)	2.8					24 400	48 500	15 000	**IRB 1220**
25.400 (1)	31.750 (1 1/4)	12.70(.500)	—					19 400	36 000	6 000	**IRB 128**
25.400 (1)	31.750 (1 1/4)	19.05(.750)	—					28 200	58 000	6 000	**IRB 1212**
25.400 (1)	31.750 (1 1/4)	25.40(1.000)	—					36 300	80 300	6 000	**IRB 1216**
25.400 (1)	33.338 (1 5/16)	12.70(.500)	3.4					10 200	13 100	15 000	**IRB 128**
25.400 (1)	33.338 (1 5/16)	15.88(.625)	3.4					15 300	22 100	15 000	—
25.400 (1)	33.338 (1 5/16)	19.05(.750)	3.4					19 300	29 700	15 000	**IRB 1212**
25.400 (1)	33.338 (1 5/16)	22.22(.875)	3.4					23 000	37 200	15 000	**IRB 1214**
25.400 (1)	33.338 (1 5/16)	25.40(1.000)	3.4	25.400	25.387	33.352	33.327	26 400	44 500	15 000	**IRB 1216**
25.400 (1)	33.338 (1 5/16)	31.75(1.250)	3.4					33 200	59 600	15 000	**IRB 1220**
25.400 (1)	33.338 (1 5/16)	38.10(1.500)	3.4					39 400	74 400	15 000	—
25.400 (1)	33.338 (1 5/16)	12.70(.500)	—					20 900	34 100	6 000	**IRB 128**
25.400 (1)	33.338 (1 5/16)	19.05(.750)	—					30 700	56 100	6 000	**IRB 1212**
25.400 (1)	33.338 (1 5/16)	25.40(1.000)	—					39 900	78 400	6 000	**IRB 1216**
26.988 (1 1/16)	33.338 (1 5/16)	15.88(.625)	2.8	26.988	26.975	33.352	33.327	11 600	19 200	14 000	—
26.988 (1 1/16)	33.338 (1 5/16)	25.40(1.000)	2.8					20 000	38 300	14 000	—

Note(1) Allowable rotational speed applies to oil lubrication. For grease lubrication, a maximum of 60% of this value is allowable.
Remark Shell Type Grease Retained Full Complement Needle Roller Bearings are provided with prepacked grease. Standard type and closed end type bearings are not provided with prepacked grease, so perform proper lubrication when using these types of bearings.

1N=0.102kgf=0.2248lbs.
1mm=0.03937inch

SHELL TYPE NEEDLE ROLLER BEARINGS

Inch Series

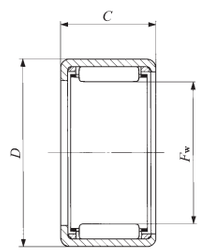


Shaft dia. 28.575 — 30.162mm

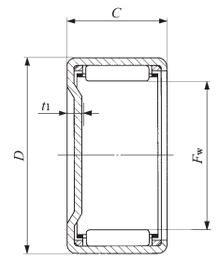
Shaft dia. mm (inch)	Identification number									
	Standard	Mass (Ref.) g	Closed end	Mass (Ref.) g	Standard	Mass (Ref.) g	Closed end	Mass (Ref.) g	Grease retained	Mass (Ref.) g
28.575 (1 1/8)	**BA 186 Z**	14.5	**BAM 186**	18.1	—	—	—	—	—	—
	BA 188 Z	19.5	**BAM 188**	23	—	—	—	—	—	—
	BA 1812 Z	29.5	**BAM 1812**	33	—	—	—	—	—	—
	BA 1816 Z	39	**BAM 1816**	42.5	—	—	—	—	—	—
	BA 1820 Z	48.5	**BAM 1820**	52	—	—	—	—	—	—
	—	—	—	—	—	—	—	—	**YB 188**	25.5
	—	—	—	—	—	—	—	—	**YB 1812**	38.5
	—	—	—	—	—	—	—	—	**YB 1816**	51.5
	—	—	—	—	**BHA 1812 Z**	45	**BHAM 1812**	49	—	—
	—	—	—	—	**BHA 1816 Z**	60	**BHAM 1816**	64	—	—
	—	—	—	—	**BHA 1818 Z**	67.5	**BHAM 1818**	71.5	—	—
	—	—	—	—	**BHA 1820 Z**	73.5	**BHAM 1820**	78	—	—
30.162 (1 3/16)	**BA 1910 Z**	32.5	**BAM 1910**	37.5	—	—	—	—	—	—
	BA 1916 Z	52	**BAM 1916**	57	—	—	—	—	—	—
	—	—	—	—	—	—	—	—	**YB 1910**	42.5

Note(1) Allowable rotational speed applies to oil lubrication. For grease lubrication, a maximum of 60% of this value is allowable.

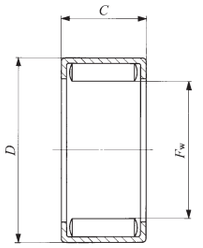
Remark Shell Type Grease Retained Full Complement Needle Roller Bearings are provided with prepacked grease. Standard type and closed end type bearings are not provided with prepacked grease, so perform proper lubrication when using these types of bearings.



BA···Z BHA···Z



BAM BHAM



YB

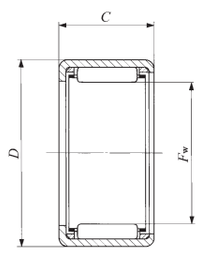
Boundary dimensions mm(inch)				Standard mounting dimensions mm				Basic dynamic load rating	Basic static load rating	Allowable rotational speed(1)	Assembled inner ring
				Shaft dia. h6		Housing bore dia. J7					
F_w	D	C	t_1 Max.	Max.	Min.	Max.	Min.	C N	C_0 N	rpm	
28.575 (1 1/8)	34.925 (1 3/8)	9.52(. 375)	2.8	28.575	28.562	34.939	34.914	6 330	8 910	13 000	—
28.575 (1 1/8)	34.925 (1 3/8)	12.70(.500)	2.8					8 680	13 400	13 000	**IRB 148**
28.575 (1 1/8)	34.925 (1 3/8)	19.05(.750)	2.8					15 000	26 900	13 000	**IRB 1412**
28.575 (1 1/8)	34.925 (1 3/8)	25.40(1.000)	2.8					20 500	40 300	13 000	**IRB 1416**
28.575 (1 1/8)	34.925 (1 3/8)	31.75(1.250)	2.8					25 700	53 900	13 000	**IRB 1420**
28.575 (1 1/8)	34.925 (1 3/8)	12.70(.500)	—					20 700	40 500	5 500	**IRB 148**
28.575 (1 1/8)	34.925 (1 3/8)	19.05(.750)	—					30 000	65 300	5 500	**IRB 1412**
28.575 (1 1/8)	34.925 (1 3/8)	25.40(1.000)	—					38 700	90 400	5 500	**IRB 1416**
28.575 (1 1/8)	38.100 (1 1/2)	19.05(.750)	3.4					22 500	32 200	13 000	**IRB 1412**
28.575 (1 1/8)	38.100 (1 1/2)	25.40(1.000)	3.4	28.575	28.562	38.114	38.089	30 900	48 600	13 000	**IRB 1416**
28.575 (1 1/8)	38.100 (1 1/2)	28.58(1.125)	3.4					34 900	56 600	13 000	—
28.575 (1 1/8)	38.100 (1 1/2)	31.75(1.250)	3.4					37 100	61 100	13 000	**IRB 1420**
30.162 (1 3/16)	38.100 (1 1/2)	15.88(.625)	2.8	30.162	30.146	38.114	38.089	15 000	22 500	12 000	—
30.162 (1 3/16)	38.100 (1 1/2)	25.40(1.000)	2.8					25 800	45 300	12 000	—
30.162 (1 3/16)	38.100 (1 1/2)	15.88(.625)	—					28 400	53 600	5 000	—

1N=0.102kgf=0.2248lbs.
1mm=0.03937inch

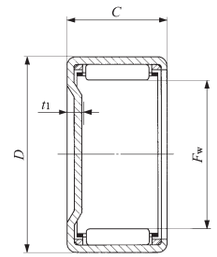
SHELL TYPE NEEDLE ROLLER BEARINGS

Inch Series

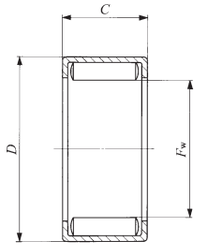




BA···Z BHA···Z



BAM BHAM



YB

Shaft dia. 31.750 — 33.338mm

Shaft dia. mm (inch)	Identification number									
	Standard	Mass (Ref.) g	Closed end	Mass (Ref.) g	Standard	Mass (Ref.) g	Closed end	Mass (Ref.) g	Grease retained	Mass (Ref.) g
31.750 (1 1/4)	**BA 208 Z**	21.5	**BAM 208**	26	—	—	—	—	—	—
	BA 2010 Z	27	**BAM 2010**	31.5	—	—	—	—	—	—
	BA 2012 Z	32.5	**BAM 2012**	37	—	—	—	—	—	—
	BA 2016 Z	43	**BAM 2016**	47.5	—	—	—	—	—	—
	BA 2020 Z	53.5	**BAM 2020**	58	—	—	—	—	—	—
	—	—	—	—	—	—	—	—	**YB 2010**	35
	—	—	—	—	—	—	—	—	**YB 2012**	42.5
	—	—	—	—	—	—	—	—	**YB 2016**	57
	—	—	—	—	—	—	—	—	**YB 2018**	64
	—	—	—	—	—	—	—	—	**YB 2020**	68
	—	—	—	—	**BHA 208 Z**	34.5	**BHAM 208**	40	—	—
	—	—	—	—	**BHA 2012 Z**	49.5	**BHAM 2012**	54.5	—	—
	—	—	—	—	**BHA 2016 Z**	66	**BHAM 2016**	71	—	—
	—	—	—	—	**BHA 2020 Z**	81.5	**BHAM 2020**	86.5	—	—
33.338 (1 5/16)	**BA 218 Z**	28.5	**BAM 218**	35	—	—	—	—	—	—
	BA 2110 Z	35.5	**BAM 2110**	41.5	—	—	—	—	—	—
	BA 2112 Z	43	**BAM 2112**	49	—	—	—	—	—	—

Boundary dimensions mm(inch)				Standard mounting dimensions mm				Basic dynamic load rating	Basic static load rating	Allowable rotational speed(1)	Assembled inner ring
				Shaft dia. h6		Housing bore dia. J7					
F_w	D	C	t_1 Max.	Max.	Min.	Max.	Min.	C N	C_0 N	rpm	
31.750 (1 1/4)	38.100 (1 1/2)	12.70(.500)	2.8					9 100	14 700	12 000	**IRB 168**
31.750 (1 1/4)	38.100 (1 1/2)	15.88(.625)	2.8					12 500	22 200	12 000	**IRB 1610**
31.750 (1 1/4)	38.100 (1 1/2)	19.05(.750)	2.8	31.750	31.734	38.114	38.089	15 700	29 600	12 000	**IRB 1612**
31.750 (1 1/4)	38.100 (1 1/2)	25.40(1.000)	2.8					21 500	44 300	12 000	**IRB 1616**
31.750 (1 1/4)	38.100 (1 1/2)	31.75(1.250)	2.8					26 900	59 200	12 000	**IRB 1620**
31.750 (1 1/4)	38.100 (1 1/2)	15.88(.625)	—					27 000	59 000	4 500	**IRB 1610**
31.750 (1 1/4)	38.100 (1 1/2)	19.05(.750)	—					31 800	72 500	4 500	**IRB 1612**
31.750 (1 1/4)	38.100 (1 1/2)	25.40(1.000)	—	31.750	31.734	38.114	38.089	40 900	100 000	4 500	**IRB 1616**
31.750 (1 1/4)	38.100 (1 1/2)	28.58(1.125)	—					45 300	114 000	4 500	—
31.750 (1 1/4)	38.100 (1 1/2)	31.75(1.250)	—					49 400	128 000	4 500	**IRB 1620**
31.750 (1 1/4)	41.275 (1 5/8)	12.70(.500)	3.4					13 700	17 600	12 000	**IRB 168**
31.750 (1 1/4)	41.275 (1 5/8)	19.05(.750)	3.4					24 100	36 400	12 000	**IRB 1612**
31.750 (1 1/4)	41.275 (1 5/8)	25.40(1.000)	3.4	31.750	31.734	41.289	41.264	33 200	55 000	12 000	**IRB 1616**
31.750 (1 1/4)	41.275 (1 5/8)	31.75(1.250)	3.4					40 000	69 600	12 000	**IRB 1620**
33.338 (1 5/16)	41.275 (1 5/8)	12.70(.500)	2.8					11 100	15 800	11 000	**IRB 168-1**
33.338 (1 5/16)	41.275 (1 5/8)	15.88(.625)	2.8	33.338	33.322	41.289	41.264	15 400	23 900	11 000	**IRB 1610-1**
33.338 (1 5/16)	41.275 (1 5/8)	19.05(.750)	2.8					19 300	32 100	11 000	**IRB 1612-1**

Note(1) Allowable rotational speed applies to oil lubrication. For grease lubrication, a maximum of 60% of this value is allowable.
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1N=0.102kgf=0.2248lbs.
1mm=0.03937inch

SHELL TYPE NEEDLE ROLLER BEARINGS

Inch Series

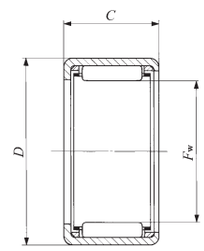


Shaft dia. 34.925 — 38.100mm

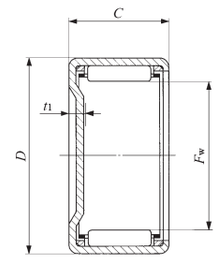
Shaft dia. mm (inch)	Identification number									
	Standard	Mass (Ref.) g	Closed end	Mass (Ref.) g	Standard	Mass (Ref.) g	Closed end	Mass (Ref.) g	Grease retained	Mass (Ref.) g
34.925 (1 3/8)	**BA 228 Z**	23.5	**BAM 228**	29	—	—	—	—	—	—
	BA 2212 Z	35.5	**BAM 2212**	41	—	—	—	—	—	—
	BA 2216 Z	47.5	**BAM 2216**	53	—	—	—	—	—	—
	BA 2220 Z	59	**BAM 2220**	64	—	—	—	—	—	—
	—	—	—	—	—	—	—	—	**YB 228**	30.5
	—	—	—	—	—	—	—	—	**YB 2212**	46
	—	—	—	—	—	—	—	—	**YB 2220**	77.5
	—	—	—	—	**BHA 228 Z**	37	**BHAM 228**	43	—	—
	—	—	—	—	**BHA 2210 Z**	44	**BHAM 2210**	50	—	—
	—	—	—	—	**BHA 2212 Z**	53	**BHAM 2212**	59	—	—
	—	—	—	—	**BHA 2216 Z**	71	**BHAM 2216**	77	—	—
	—	—	—	—	**BHA 2220 Z**	87	**BHAM 2220**	98.5	—	—
38.100 (1 1/2)	**BA 248 Z**	38.5	**BAM 248**	47.5	—	—	—	—	—	—
	BA 2410 Z	48.5	**BAM 2410**	57.5	—	—	—	—	—	—
	BA 2412 Z	58.5	**BAM 2412**	67.5	—	—	—	—	—	—
	BA 2414 Z	69	**BAM 2414**	78	—	—	—	—	—	—
	BA 2416 Z	79	**BAM 2416**	88	—	—	—	—	—	—
	BA 2420 Z	97.5	**BAM 2420**	106	—	—	—	—	—	—
	—	—	—	—	—	—	—	—	**YB 246**	38
	—	—	—	—	—	—	—	—	**YB 248**	51.5
	—	—	—	—	—	—	—	—	**YB 2414**	91
	—	—	—	—	—	—	—	—	**YB 2416**	105
	—	—	—	—	—	—	—	—	**YB 2420**	131

Note(1) Allowable rotational speed applies to oil lubrication. For grease lubrication, a maximum of 60% of this value is allowable.

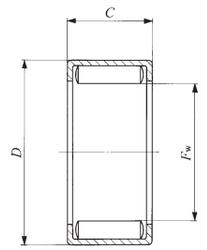
Remark Shell Type Grease Retained Full Complement Needle Roller Bearings are provided with prepacked grease. Standard type and closed end type bearings are not provided with prepacked grease, so perform proper lubrication when using these types of bearings.



BA···Z BHA···Z



BAM BHAM



YB

Boundary dimensions mm(inch)				Standard mounting dimensions mm				Basic dynamic load rating	Basic static load rating	Allowable rotational speed(1)	Assembled inner ring
				Shaft dia. h6		Housing bore dia. J7					
F_w	D	C	t_1 Max.	Max.	Min.	Max.	Min.	C N	C_0 N	rpm	
34.925 (1 3/8)	41.275 (1 5/8)	12.70(.500)	2.8					9 770	16 600	10 000	**IRB 188**
34.925 (1 3/8)	41.275 (1 5/8)	19.05(.750)	2.8					16 900	33 500	10 000	**IRB 1812**
34.925 (1 3/8)	41.275 (1 5/8)	25.40(1.000)	2.8					23 100	50 200	10 000	**IRB 1816**
34.925 (1 3/8)	41.275 (1 5/8)	31.75(1.250)	2.8	34.925	34.909	41.289	41.264	28 900	67 100	10 000	**IRB 1820**
34.925 (1 3/8)	41.275 (1 5/8)	12.70(.500)	—					23 000	49 500	4 500	**IRB 188**
34.925 (1 3/8)	41.275 (1 5/8)	19.05(.750)	—					33 400	79 800	4 500	**IRB 1812**
34.925 (1 3/8)	41.275 (1 5/8)	31.75(1.250)	—					52 000	141 000	4 500	**IRB 1820**
34.925 (1 3/8)	44.450 (1 3/4)	12.70(.500)	3.4					14 100	18 800	10 000	**IRB 188**
34.925 (1 3/8)	44.450 (1 3/4)	15.88(.625)	3.4					19 700	28 800	10 000	—
34.925 (1 3/8)	44.450 (1 3/4)	19.05(.750)	3.4	34.925	34.909	44.464	44.439	24 800	38 800	10 000	**IRB 1812**
34.925 (1 3/8)	44.450 (1 3/4)	25.40(1.000)	3.4					34 100	58 400	10 000	**IRB 1816**
34.925 (1 3/8)	44.450 (1 3/4)	31.75(1.250)	3.4					41 200	74 200	10 000	**IRB 1820**
38.100 (1 1/2)	47.625 (1 7/8)	12.70(.500)	2.8					12 900	17 900	9 000	—
38.100 (1 1/2)	47.625 (1 7/8)	15.88(.625)	2.8					17 800	27 100	9 000	**IRB 2010**
38.100 (1 1/2)	47.625 (1 7/8)	19.05(.750)	2.8	38.100	38.084	47.639	47.614	22 500	36 600	9 000	—
38.100 (1 1/2)	47.625 (1 7/8)	22.22(.875)	2.8					26 700	45 600	9 000	**IRB 2014**
38.100 (1 1/2)	47.625 (1 7/8)	25.40(1.000)	2.8					31 100	55 400	9 000	**IRB 2016**
38.100 (1 1/2)	47.625 (1 7/8)	31.75(1.250)	2.8					39 000	74 200	9 000	**IRB 2020**
38.100 (1 1/2)	47.625 (1 7/8)	9.52(.375)	—					21 000	34 100	4 000	—
38.100 (1 1/2)	47.625 (1 7/8)	12.70(.500)	—					28 700	50 900	4 000	—
38.100 (1 1/2)	47.625 (1 7/8)	22.22(.875)	—	38.100	38.084	47.639	47.614	48 900	101 000	4 000	**IRB 2014**
38.100 (1 1/2)	47.625 (1 7/8)	25.40(1.000)	—					55 100	118 000	4 000	**IRB 2016**
38.100 (1 1/2)	47.625 (1 7/8)	31.75(1.250)	—					66 800	151 000	4 000	**IRB 2020**

1N=0.102kgf=0.2248lbs.
1mm=0.03937inch

SHELL TYPE NEEDLE ROLLER BEARINGS

Inch Series



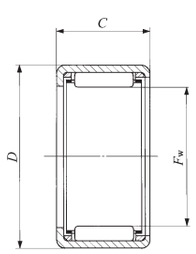
Shaft dia. 41.275 — 52.388mm

Shaft dia. mm (inch)	Identification number									
	Standard	Mass (Ref.) g	Closed end	Mass (Ref.) g	Standard	Mass (Ref.) g	Closed end	Mass (Ref.) g	Grease retained	Mass (Ref.) g
41.275 (1 5/8)	**BA 268 Z**	41	**BAM 268**	51.5	—	—	—	—	—	—
	BA 2610 Z	52	**BAM 2610**	62.5	—	—	—	—	—	—
	BA 2616 Z	85	**BAM 2616**	95.5	—	—	—	—	—	—
	BA 2620 Z	105	**BAM 2620**	115	—	—	—	—	—	—
	—	—	—	—	—	—	—	—	**YB 2610**	69
44.450 (1 3/4)	**BA 2812 Z**	67.5	**BAM 2812**	79.5	—	—	—	—	—	—
	BA 2816 Z	91	**BAM 2816**	103	—	—	—	—	—	—
	BA 2820 Z	112	**BAM 2820**	125	—	—	—	—	—	—
	BA 2824 Z	136	**BAM 2824**	148	—	—	—	—	—	—
	—	—	—	—	—	—	—	—	**YB 2816**	119
	—	—	—	—	**BHA 2824 Z**	195	**BHAM 2824**	210	—	—
47.625 (1 7/8)	**BA 308 Z**	47.5	**BAM 308**	61	—	—	—	—	—	—
	BA 3010 Z	60	**BAM 3010**	74	—	—	—	—	—	—
	BA 3012 Z	72.5	**BAM 3012**	86.5	—	—	—	—	—	—
	BA 3016 Z	97.5	**BAM 3016**	112	—	—	—	—	—	—
	—	—	—	—	—	—	—	—	**YB 3012**	95
50.800 (2)	**BA 328 Z**	50	**BAM 328**	66	—	—	—	—	—	—
	BA 3216 Z	104	**BAM 3216**	119	—	—	—	—	—	—
	BA 3220 Z	128	**BAM 3220**	144	—	—	—	—	—	—
	BA 3224 Z	155	**BAM 3224**	170	—	—	—	—	—	—
	BAW3228Z	180	**BAMW3228**	196	—	—	—	—	—	—
	—	—	—	—	—	—	—	—	**YB 3216**	130
52.388 (2 1/16)	—	—	—	—	**BHA 3312 Z**	104	**BHAM 3312**	122	—	—
	—	—	—	—	**BHA 3316 Z**	139	**BHAM 3316**	157	—	—
	—	—	—	—	**BHA 3324 Z**	205	**BHAM 3324**	225	—	—

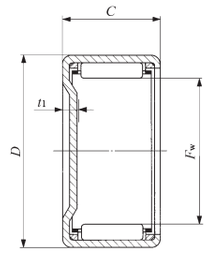
Note(1) Allowable rotational speed applies to oil lubrication. For grease lubrication, a maximum of 60% of this value is allowable.

Remarks1. "W" in the identification number indicates that rolling elements are arranged in double rows.

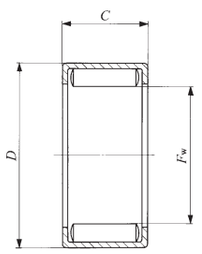
2. Shell Type Grease Retained Full Complement Needle Roller Bearings are provided with prepacked grease. Standard type and closed end type bearings are not provided with prepacked grease, so perform proper lubrication when using these types of bearings.



BA···Z BHA···Z



BAM BHAM



YB

Boundary dimensions mm(inch)				Standard mounting dimensions mm				Basic dynamic load rating	Basic static load rating	Allowable rotational speed(1)	Assembled inner ring
				Shaft dia. h6		Housing bore dia. J7					
F_w	D	C	t_1 Max.	Max.	Min.	Max.	Min.	C N	C_0 N	rpm	
41.275 (1 5/8)	50.800 (2)	12.70(.500)	2.8					13 700	19 800	8 000	—
41.275 (1 5/8)	50.800 (2)	15.88(.625)	2.8					18 900	30 000	8 000	**IRB 2210**
41.275 (1 5/8)	50.800 (2)	25.40(1.000)	2.8	41.275	41.259	50.818	50.788	33 000	61 400	8 000	—
41.275 (1 5/8)	50.800 (2)	31.75(1.250)	2.8					41 400	82 100	8 000	**IRB 2220**
41.275 (1 5/8)	50.800 (2)	15.88(.625)	—					37 000	71 700	3 500	**IRB 2210**
44.450 (1 3/4)	53.975 (2 3/8)	19.05(.750)	2.8					25 200	44 500	7 500	**IRB 2412**
44.450 (1 3/4)	53.975 (2 3/8)	25.40(1.000)	2.8					34 800	67 400	7 500	**IRB 2416**
44.450 (1 3/4)	53.975 (2 3/8)	31.75(1.250)	2.8	44.450	44.434	53.993	53.963	43 600	90 200	7 500	—
44.450 (1 3/4)	53.975 (2 3/8)	38.10(1.500)	2.8					52 000	113 000	7 500	**IRB 2424**
44.450 (1 3/4)	53.975 (2 3/8)	25.40(1.000)	—					59 500	136 000	3 500	**IRB 2416**
44.450 (1 3/4)	57.150 (2 1/4)	38.10(1.500)	3.4	44.450	44.434	57.168	57.138	72 200	135 000	7 500	**IRB 2424**
47.625 (1 7/8)	57.150 (2 1/4)	12.70(.500)	2.8					14 700	22 800	7 000	**IRB 248-1**
47.625 (1 7/8)	57.150 (2 1/4)	15.88(.625)	2.8					20 300	34 500	7 000	**IRB 2410-1**
47.625 (1 7/8)	57.150 (2 1/4)	19.05(.750)	2.8	47.625	47.609	57.168	57.138	25 700	46 700	7 000	—
47.625 (1 7/8)	57.150 (2 1/4)	25.40(1.000)	2.8					35 400	70 600	7 000	—
47.625 (1 7/8)	57.150 (2 1/4)	19.05(.750)	—					47 800	105 000	3 000	—
50.800 (2)	60.325 (2 3/8)	12.70(.500)	2.8					15 400	24 700	6 000	—
50.800 (2)	60.325 (2 3/8)	25.40(1.000)	2.8					37 100	76 500	6 000	**IRB 2616**
50.800 (2)	60.325 (2 3/8)	31.75(1.250)	2.8					46 600	102 000	6 000	**IRB 2720**
50.800 (2)	60.325 (2 3/8)	38.10(1.500)	2.8	50.800	50.781	60.343	60.313	55 500	128 000	6 000	—
50.800 (2)	60.325 (2 3/8)	44.45(1.750)	2.8					57 900	136 000	6 000	**IRB 2628**
50.800 (2)	60.325 (2 3/8)	25.40(1.000)	—					64 100	156 000	2 500	**IRB 2616**
52.388 (2 1/16)	64.294 (2 17/32)	19.05(.750)	3.4					36 400	62 100	6 000	—
52.388 (2 1/16)	64.294 (2 17/32)	25.40(1.000)	3.4	52.388	52.369	64.312	64.282	50 600	94 700	6 000	—
52.388 (2 1/16)	64.294 (2 17/32)	38.10(1.500)	3.4					73 900	154 000	6 000	—

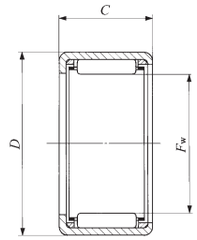
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1mm=0.03937inch

SHELL TYPE NEEDLE ROLLER BEARINGS

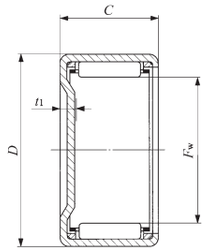
Inch Series







BA···Z



BAM

Shaft dia. 53.975 — 69.850mm

Shaft dia. mm (inch)	Identification number									
	Standard	Mass (Ref.) g	Closed end	Mass (Ref.) g	Standard	Mass (Ref.) g	Closed end	Mass (Ref.) g	Grease retained	Mass (Ref.) g
53.975 (2 1/8)	**BA 348 Z**	53	**BAM 348**	70.5	—	—	—	—	—	—
	BA 3416 Z	109	**BAM 3416**	127	—	—	—	—	—	—
	BA 3424 Z	162	**BAM 3424**	180	—	—	—	—	—	—
57.150 (2 1/4)	**BA 3612 Z**	85.5	**BAM 3612**	105	—	—	—	—	—	—
	BA 3616 Z	115	**BAM 3616**	135	—	—	—	—	—	—
	BA 3620 Z	143	**BAM 3620**	163	—	—	—	—	—	—
	BA 3624 Z	172	**BAM 3624**	192	—	—	—	—	—	—
66.675 (2 5/8)	**BA 4216 Z**	133	**BAM 4216**	161	—	—	—	—	—	—
69.850 (2 3/4)	**BA 4410 Z**	85.5	**BAM 4410**	115	—	—	—	—	—	—
	BA 4412 Z	103	**BAM 4412**	133	—	—	—	—	—	—
	BA 4416 Z	139	**BAM 4416**	169	—	—	—	—	—	—
	BA 4420 Z	173	**BAM 4420**	205	—	—	—	—	—	—

Boundary dimensions mm(inch)				Standard mounting dimensions mm				Basic dynamic load rating	Basic static load rating	Allowable rotational speed(1)	Assembled inner ring
				Shaft dia. h6		Housing bore dia. J7					
F_w	D	C	t_1 Max.	Max.	Min.	Max.	Min.	C N	C_0 N	rpm	
53.975 (2 1/8)	63.500 (2 1/2)	12.70(.500)	2.8					16 100	26 600	5 500	—
53.975 (2 1/8)	63.500 (2 1/2)	25.40(1.000)	2.8	53.975	53.956	63.518	63.488	38 700	82 500	5 500	**IRB 3016**
53.975 (2 1/8)	63.500 (2 1/2)	38.10(1.500)	2.8					57 900	138 000	5 500	**IRB 3024**
57.150 (2 1/4)	66.675 (2 5/8)	19.05(.750)	2.8					28 500	56 700	5 000	—
57.150 (2 1/4)	66.675 (2 5/8)	25.40(1.000)	2.8					39 300	85 700	5 000	—
57.150 (2 1/4)	66.675 (2 5/8)	31.75(1.250)	2.8	57.150	57.131	66.693	66.663	49 400	115 000	5 000	—
57.150 (2 1/4)	66.675 (2 5/8)	38.10(1.500)	2.8					58 800	144 000	5 000	—
66.675 (2 5/8)	76.200 (3)	25.40(1.000)	2.8	66.675	66.656	76.218	76.188	42 000	97 900	4 000	**IRB 3616**
69.850 (2 3/4)	79.375 (3 1/8)	15.88(.625)	2.8					25 000	50 800	3 500	—
69.850 (2 3/4)	79.375 (3 1/8)	19.05(.750)	2.8					31 500	68 700	3 500	—
69.850 (2 3/4)	79.375 (3 1/8)	25.40(1.000)	2.8	69.850	69.831	79.393	79.363	43 500	104 000	3 500	**IRB 4016**
69.850 (2 3/4)	79.375 (3 1/8)	31.75(1.250)	2.8					54 600	139 000	3 500	**IRB 4020**

Note(1) Allowable rotational speed applies to oil lubrication. For grease lubrication, a maximum of 60% of this value is allowable.
Remark Shell Type Grease Retained Full Complement Needle Roller Bearings are provided with prepacked grease. Standard type and closed end type bearings are not provided with prepacked grease, so perform proper lubrication when using these types of bearings.

1N=0.102kgf=0.2248lbs.
1mm=0.03937inch

SHELL TYPE NEEDLE ROLLER BEARINGS

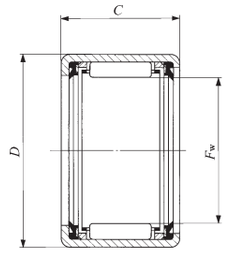
With seals



Shaft dia. 12 — 50mm

Shaft dia. mm	Identification number	Mass (Ref.) g	Boundary dimensions mm F_w	D	C	Standard mounting dimensions mm Shaft dia. h6 Max.	Shaft dia. h6 Min.	Housing bore dia. N7 Max.	Housing bore dia. N7 Min.
12	**TLA 1216 UU**	11.7	12	18	16	12.000	11.989	17.995	17.977
14	**TLA 1416 UU**	13.3	14	20	16	14.000	13.989	19.993	19.972
15	**TLA 1516 UU**	14	15	21	16	15.000	14.989	20.993	20.972
16	**TLA 1616 UU**	14.8	16	22	16	16.000	15.989	21.993	21.972
18	**TLA 1816 UU**	16.3	18	24	16	18.000	17.989	23.993	23.972
20	**TLA 2016 UU** **TLA 2020 UU**	17.8 22.5	20 20	26 26	16 20	20.000	19.987	25.993	25.972
22	**TLA 2216 UU** **TLA 2220 UU**	19.4 25	22 22	28 28	16 20	22.000	21.987	27.993	27.972
25	**TLA 2516 UU** **TLA 2520 UU**	26 33	25 25	32 32	16 20	25.000	24.987	31.992	31.967
28	**TLA 2820 UU**	36.5	28	35	20	28.000	27.987	34.992	34.967
30	**TLA 3016 UU** **TLA 3020 UU**	30.5 39	30 30	37 37	16 20	30.000	29.987	36.992	36.967
35	**TLA 3516 UU** **TLA 3520 UU**	35 45	35 35	42 42	16 20	35.000	34.984	41.992	41.967
40	**TLA 4016 UU** **TLA 4020 UU**	39.5 50.5	40 40	47 47	16 20	40.000	39.984	46.992	46.967
45	**TLA 4520 UU**	56	45	52	20	45.000	44.984	51.991	51.961
50	**TLA 5026 UU**	89	50	58	26	50.000	49.984	57.991	57.961

Note(1) Allowable rotational speed applies to grease lubrication.
Remark The type with seals is provided with prepacked grease.



TLA···UU

Basic dynamic load rating C N	Basic static load rating C_0 N	Allowable rotational speed(1) rpm
6 420	7 490	14 000
7 080	8 840	12 000
7 380	9 520	11 000
7 670	10 200	11 000
8 230	11 500	9 000
8 740 11 100	12 900 17 500	9 000 9 000
9 230 11 700	14 300 19 300	8 000 8 000
9 440 12 800	13 900 20 500	7 000 7 000
13 800	23 500	6 000
10 400 14 100	16 600 24 500	5 500 5 500
11 600 15 700	20 000 29 600	5 000 5 000
12 400 16 700	22 800 33 700	4 500 4 500
17 800	37 800	4 000
28 800	64 100	3 500

1N=0.102kgf=0.2248lbs.
1mm=0.03937inch

NEEDLE ROLLER CAGES FOR GENERAL USAGE



Structure and Features

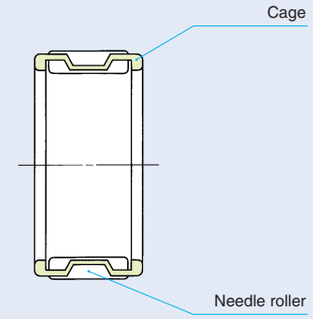
IKO Needle Roller Cages for General Usage are bearings which display excellent rotational performance. Needle rollers with extremely small dimensional variations in diameter are incorporated and retained in their specially shaped cages with high rigidity and accuracy, which precisely guide the needle rollers.

When combined with shafts and housing bores that are heat treated and accurately ground as raceway surfaces, Needle Roller Cages for General Usage are particularly useful in small spaces.

In addition, since they are lightweight and have high rigidity as well as a large lubricant holding capacity, they can withstand severe operating conditions such as high speed rotation and shock loads, and they are used in a wide range of applications.

Structures of Needle Roller Cages for General Usage

KT



KTW

